

Lab 3: Small-Scale Fading

Objectives

- Observe small-scale fading of a radio frequency channel
- Analyze spatial variations and statistics of real fading
- Compare real fading to Rayleigh fading model

Equipment needed

- ISM transmitter (915 MHz)
- ISM field strength meter (915 MHz)
- Computer running Matlab

Procedure

1. Place the ISM transmitter on a metal stool outside the RF lab (EME240). The door to the lab should be open, and the metal stool with transmitter should be placed outside the lab and just to the side of the doorway, so that there is no line of sight between the transmitter and the center of the lab.
2. Using an ISM field strength meter, measure the received power on a square grid at two-inch intervals. Measure eight intervals in each dimension for a total of 64 field samples. Record your measurements in a square array keeping track of the spatial location corresponding to each measurement. Try to have everyone remain in the same place as you make your measurements.

Analysis

1. Put your measurements into Matlab. Produce contour plots of field strength in dBm vs. location. (The "contour" command will do this.)
2. Convert the dBm values into rms voltages across 50Ω using $P(\text{dBm}) = 10 \log \left(1000 \frac{V^2}{50} \right)$.

Make histograms of these voltages (Matlab's "hist" command will do this, but note that it works on a vector, not a matrix). Plot the theoretical Rayleigh distribution discussed in Lecture 8. Scale it so that it more-or-less matches the observed data.

Write Up

- Briefly describe how you conducted your experiment.
- Explain in detail how you analyzed your data.
- Present your results, including code and plots.

- Write a conclusion discussing what you learned from this lab. Convince me that you've really spent some time thinking about this experiment and how it relates to what we've learned in lecture!